

GENERAL STRUCTURAL NOTES:

1. THE GOVERNING BUILDING CODE IS THE MINNESOTA STATE BUILDING CODE 2020 EDITION AS APPROVED AND AMENDED BY THE CITY OF ST CLOUD, MN.
2. CONTRACT DOCUMENTS INCLUDE THE STRUCTURAL DRAWINGS AND SPECIFICATIONS, BUT DO NOT INCLUDE SHOP DRAWINGS, VENDOR DRAWINGS, OR OTHER SUBMITTALS BY THE CONTRACTOR.
3. CONTRACTOR SHALL CROSS VERIFY ALL CONTRACT DOCUMENTS, ELEVATIONS, DIMENSIONS, AND EXISTING CONDITIONS PRIOR TO STARTING WORK. DISCREPANCIES OR CONFLICTS SHALL BE NOTED TO THE EOR IMMEDIATELY FOR REMEDIATION. SPECIFIC NOTES AND DETAILS SHALL PRESIDE OVER GENERAL NOTES AND SPECIFICATIONS.
4. THE DIMENSIONS, LOCATIONS, AND DETAILS SHOWN ON THE DRAWINGS ARE BASED ON THE BEST AVAILABLE INFORMATION AT THE TIME OF THE DRAWINGS BEING ISSUED. DEVIATIONS WHICH ARE NECESSARY OR WHICH CONFLICT SHALL BE REPORTED TO THE EOR. CONTRACTOR SHALL HAVE FULL RESPONSIBILITY FOR DEVIATIONS NOT APPROVED BY THE EOR.
5. COSTS OF ADDITIONAL DESIGN WORK DUE TO THE SELECTION OF AN OPTION OR DUE TO ERRORS OR OMISSION IN CONSTRUCTION SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR.
6. THE CONTRACTOR HAS SOLE RESPONSIBILITY FOR THE MEANS, METHODS, TECHNIQUES, SEQUENCES, AND PROCEDURES OF CONSTRUCTION. THE STRUCTURAL DRAWINGS REFLECT THE COMPLETED STRUCTURE, BRACING, SHORING, AND PROTECTION DURING CONSTRUCTION IS THE RESPONSIBILITY OF THE CONTRACTOR. THE STRUCTURE SHALL NOT BE LOADED WITH CONSTRUCTION MATERIALS AND EQUIPMENT THAT EXCEEDS THE DESIGN LOADS.
7. PENETRATIONS NOT SHOWN ON THE DRAWINGS MUST BE APPROVED BY THE EOR BEFORE PLACING THROUGH STRUCTURAL ELEMENTS. CONTRACTOR SHALL PROVIDE A CAST-IN SLEEVE FOR ALL HORIZONTAL ELEMENTS THAT EXTEND THROUGH FOOTING AND FOUNDATION WALL, SUCH AS DRAIN TILE, CONDUIT, PIPING, ETC. COORDINATE SLEEVES WITH EOR. SEE MECHANICAL, ELECTRICAL, AND ARCHITECTURAL DRAWINGS FOR ALL PENETRATIONS AND EMBEDDED ITEMS NOT SHOWN ON THE STRUCTURAL DRAWINGS.
8. CONTRACTOR TO VERIFY ALL WEIGHTS, LOCATIONS & DIMENSIONS OF MECH. EQUIPMENT SHOWN AND NOTIFY THE EOR OF ANY DISCREPANCIES. COORDINATE THIS INFORMATION WITH ALL NECESSARY INDIVIDUALS.
9. PERIODIC SITE OBSERVATION BY REPRESENTATIVES OF SANDMAN STRUCTURAL ENGINEERS IS SOLELY FOR THE PURPOSE OF DETERMINING IF THE WORK OF THE CONTRACTOR IS PROCEEDING IN GENERAL ACCORDANCE WITH THE STRUCTURAL CONTRACT DRAWINGS. A LIMITED SITE OBSERVATION SHOULD NOT BE CONSTRUED AS EXHAUSTIVE OR CONTINUOUS TO CHECK THE QUALITY OR QUANTITY OF THE WORK, BUT RATHER PERIODIC IN AN EFFORT TO GUARD THE OWNER AGAINST DEFECTS OR DEFICIENCIES IN THE WORK OF THE CONTRACTOR. ALL MATERIALS AND WORKMANSHIP ARE SUBJECT TO THE REVIEW OF THE EOR.
10. SEE THE FOLLOWING DETAILS FOR STANDARD DETAILS:

CIP CONCRETE & FOUNDATION: 1/3S01
STRUCTURAL STEEL: 1/3430
CMU: 1/3322
WOOD FRAMING: 1/3410

SHOP DRAWING & DEFERRED DESIGN SUBMITTAL NOTES:

1. CONTRACTOR TO FURNISH COMPLETE AND DETAILED SHOP DRAWING & SUBMITTALS FOR REVIEW AND APPROVAL BY THE EOR. THE FOLLOWING ARE REQUIRED FOR THIS PROJECT.

SHOP DRAWING OR SUBMITTAL	NOTES/COMMENTS
CONCRETE MIX DESIGNS	BY 3RD PARTY TESTING AGENCY
CONCRETE REINFORCEMENT	-
STRUCTURAL & MISC. STEEL	-
CMU, MORTAR, & GROUT MATERIAL SPECS	-
CMU REINFORCEMENT	-
PRECAST/PRESTRESSED CONCRETE	DEFERRED SUBMITTAL: SEE NOTE #4
LUMBER MATERIAL GRADE & DOVEL-TYPE FASTENERS	-
METAL PLATE CONNECTED WOOD TRUSSES	DEFERRED SUBMITTAL: SEE NOTE #4
ALUMINUM FRAMING ASSEMBLIES	DEFERRED SUBMITTAL: SEE NOTE #4
HANDRAILS AND GUARDRAILS	DEFERRED SUBMITTAL: SEE NOTE #4
ELEVATOR	PROVIDE LOADS @ STRUCTURAL SYSTEM
SIMPSON STRONG-TIE ANCHOR TIEDOWN SYSTEM	-
2. CONTRACT DRAWINGS SHALL NOT BE REPRODUCED FOR USE AS SHOP DRAWINGS. ALL SHOP DRAWINGS MUST BEAR THE REVIEW STAMP OF THE CONTRACTOR BEFORE THEY ARE REVIEWED BY THE EOR.
3. SHOP DRAWINGS SHALL SHOW ALL FIELD DETAILS AND ADDITIONAL INFORMATION NEEDED FOR THE CONTRACTOR TO CONSTRUCT THE BUILDING PER THE CONTRACT DOCUMENTS.
4. STRUCTURAL COMPONENTS/SYSTEMS DESIGNATED AS A "DEFERRED SUBMITTAL" OR AS "DELEGATED DESIGN," "DESIGNED BY OTHERS," OR "PRE-ENGINEERING" MUST INCLUDE A CALCULATION PACKAGE THAT IS STAMPED AND SIGNED BY A REGISTERED PROFESSIONAL ENGINEER, LICENSED IN THE STATE THE PROJECT WILL BE CONSTRUCTED, PRIOR TO SUBMITTAL FOR APPROVAL TO THE EOR.
5. SHOP DRAWINGS SHALL BE AVAILABLE ON THE JOB SITE DURING TIMES OF INSPECTION AND SHALL BE CLEARLY INDICATED THAT THEY HAVE BEEN REVIEWED AND APPROVED BY THE EOR.
6. REVIEW OF SUBMITTALS AND SHOP DRAWINGS BY THE EOR DOES NOT RELIEVE THE CONTRACTOR OF THE SOLE RESPONSIBILITY TO REVIEW AND CHECK ALL SUBMITTALS AND SHOP DRAWINGS BEFORE SUBMITTING TO THE STRUCTURAL ENGINEER. THE CONTRACTOR REMAINS SOLELY RESPONSIBLE FOR THE ERRORS AND OMISSIONS ASSOCIATED WITH THE PREPARATION OF SHOP DRAWINGS AS THEY PERTAIN TO MEMBER SIZES, DETAILS, AND DIMENSIONS SPECIFIED IN THE CONTRACT DOCUMENTS.
7. ALL HANDRAIL AND GUARDRAIL SYSTEMS SHALL BE DESIGNED TO RESIST ASCE-7 CHAPTER 4 DESIGN LOADING CRITERIA AND TO TRANSFER THESE LOADS THROUGH THE SUPPORTS TO THE STRUCTURE.

DESIGN CRITERIA/DESIGN LOADS:

SNOW LOAD DESIGN CRITERIA				
GROUND SNOW LOAD		Pg	50 PSF	-
SNOW IMPORTANCE FACTOR		Is	1.0	-
EXPOSURE FACTOR		Ce	1.0	-
THERMAL FACTOR		Ct	1.1	TYPICAL
THERMAL FACTOR		Ct	1.2	CANOPY
SLOPED ROOF FACTOR		Cs	1.0	TYPICAL/CANOPY
ROOF DESIGN LOADS				
LOAD TYPE		NOTATION	LOAD	NOTES
SNOW LOAD		S	38.5 PSF	TYPICAL (Ps)
SNOW LOAD		S	42 PSF	CANOPY (Pi)
LIVE LOAD		RLL	20 PSF	-
DEAD LOAD		D	20 PSF	D TC = 12 PSF D BC = 8 PSF
FLOOR DESIGN LOADS				
FLOOR	DESCRIPTION	DEAD LOAD (D)	LIVE LOAD (L)	NOTES
1ST	TYP RESIDENTIAL	95 PSF*/135 PSF**	40 PSF***	-
1ST	STAIRS/LANDINGS/ CORRIDOR/ PUBLIC AREAS / COMMUNITY SPACE	95 PSF*/135 PSF**	100 PSF	-
1ST	ELECT/MECH/ TRASH/STORAGE	95 PSF*/135 PSF**	80 PSF***	-
2ND-3RD	TYP RESIDENTIAL/ CORRIDORS	26 PSF	40 PSF***	D TC = 18 PSF D BC = 8 PSF
2ND-3RD	STAIRS/LANDINGS/ PUBLIC AREAS / COMMUNITY SPACE	26 PSF	100 PSF	D TC = 18 PSF D BC = 8 PSF
2ND-3RD	ELECT/MECH/ TRASH/STORAGE	26 PSF	80 PSF***	D TC = 18 PSF D BC = 8 PSF
1ST-3RD	BALCONIES	SELF-WEIGHT	60 PSF	TOP LEVEL BALCONIES DESIGN FOR: S = 105 PSF WIND UPLIFT = 15 PSF (SERVICE)
*D BREAKDOWN (8" H.C.) = 60 PSF H.C. + 25 PSF CONCRETE TOPPING + 10 PSF S.I. **D BREAKDOWN (12" H.C.) = 100 PSF H.C. + 25 PSF CONCRETE TOPPING + 10 PSF S.I. ***DENOTES LIVE LOAD IS REDUCIBLE PER ASCE 7				
WIND LOAD DESIGN CRITERIA				
ULT. DESIGN WIND SPEED		V-ult	109 MPH	-
NOMINAL DESIGN WIND SPEED		V-asd	84.4 MPH	-
RISK CATEGORY		-	II	-
EXPOSURE CATEGORY		-	C	-
INTERNAL PRESSURE COEF.		GCPi	+/- 0.18	-
C & C BASE PRESSURE		qh - ult	27.1 PSF	-
END ZONE WIDTH		a	6.5 FT	-
EQUIVALENT LATERAL EARTH PRESSURES				
SOIL		TYPE	PRESSURE	NOTES
PER GEOTECH REPORT		AT REST	55 PCF	USED FOR BASEMENT WALL DESIGN
REFER TO GEOTECHNICAL REPORT FOR BACKFILLING REQUIREMENTS. SOIL RETAINING/BASEMENT WALLS ON PROJECT HAVE NOT BEEN DESIGNED TO WITHSTAND HYDROSTATIC PRESSURE. PROVIDE DRAIN TILE SYSTEM & DRAIN PATH ALONG BACKSIDE OF WALL TO RELIEVE HYDROSTATIC PRESSURE. REFER TO GEOTECHNICAL REPORT FOR ADDITIONAL REQUIREMENTS.				
ESTIMATED WOOD BUILDING SHRINKAGE				
WOOD LEVEL		LEVEL SHRINKAGE		CUMULATIVE SHRINKAGE
3RD		.164"		.648"
2ND		.239"		.484"
1ST		.245"		.245"
ESTIMATED SHRINKAGE VALUES ASSUME ALL MEMBERS HAVE A MOISTURE CONTENT OF 19% AT THE TIME OF INSTALLATION AND 10% IN FINAL STATE.				
THESE VALUES HAVE BEEN PROVIDED FOR CONSIDERATION IN DESIGN AND DETAILING OF BUILDING SYSTEMS SENSATIVE TO SHRINKAGE.				
INSTALLATION OF MEP SYSTEMS ARE TO TAKE IN CONSIDERATION THE ESTIMATED SHIRNKAGE OF THE WOOD STRUCTURE.				
G.C./OWNER HAS CHOSE TO USE CMJ SHAFTS THROUGH WOOD STRUCTURE. DUE TO DIFFERENTIAL SHIRNKAGE BETWEEN MATERIALS, FINISHES @ INTERFACE MAY REQUIRE REPETATIVE MAINTNANCE.				

FOUNDATION NOTES:

1. FOOTINGS ARE DESIGNED FOR A NET ALLOWABLE SOIL BEARING PRESSURE OF 3000PSF FOR STRIP FOOTINGS AND 3000PSF FOR PAD FOOTINGS. THE ASSUMED ALLOWABLE SOIL BEARING PRESSURE SHALL BE VERIFIED BY A LICENSED GEOTECHNICAL ENGINEER AT TIME OF CONSTRUCTION.
2. ALL FILL UPON WHICH FOUNDATION ELEMENTS BEAR OR RELY UPON FOR CONFINING RESISTANCE AND/OR LATERAL SUPPORT, SHALL BE DESIGNED, SPECIFIED, AND TESTED BY A QUALIFIED GEOTECHNICAL ENGINEER. THESE SITUATIONS INCLUDE, BUT ARE NOT LIMITED TO, SITE GRADING, OVER EXCAVATION FILL, UTILITY TRENCHES, ETC.
3. TEMPORARY AND PERMANENT SLOPE STABILITY OF UNDERLYING SOILS SUPPORTING THE CONSTRUCTION AND STRUCTURE IS THE RESPONSIBILITY OF A QUALIFIED GEOTECHNICAL ENGINEER. DESIGN OF THE ABOVE AND BELOW GRADE STRUCTURAL ELEMENTS AND SYSTEMS INDICATED ON THE STRUCTURAL DRAWINGS ASSUME THAT SOPE STABILITY CONCERNS DO NOT EXIST. DESIGN AND SERVICEABILITY CONSIDERATION HAVE NOT BEEN INCORPORATED FOR DIFFERENTIAL VERTICAL AND/OR LATERAL DISPLACEMENT OF THE SOILS SUPPORTING THE STRUCTURE DUE TO SLOPE INSTABILITY.
4. UNLESS SPECIFICALLY PRESCRIBED IN A GEOTECHNICAL REPORT, BACKFILL ABOVE THE BOTTOM OF FOOTING ELEVATION AGAINST FOUNDATION WALL OR RETAINING WALL, SYSTEMS SHALL BE COMPACTED TO A MINIMUM OF 95% OF THE STANDARD PROCTOR MAX DENSITY (ASTM D698-00A). MOISTURE CONTENT AT THE TIME OF COMPACTION SHOULD BE -1% +3% OF OPTIMUM. GRADATION OF FILL MATERIAL SHALL HAVE LESS THAN 12% PASSING THE #200 SIEVE.
5. PROTECT FOOTING EXCAVATIONS FROM WATER, MOISTURE, OR FROST INFILTRATION. PRIOR TO PLACEMENT OF FOOTING CONCRETE, CLEAN FOOTING EXCAVATIONS OF SNOW, WATER, MUD, DIRT, AND DEBRIS. DO NOT PLACE FOOTINGS OR BACKFILL ON FROZEN SUB GRADE.
6. FROST COVER FOR FOOTINGS SHALL BE PER RECOMMENDATIONS OF THE GEOTECHNICAL REPORT. SEE PLAN FOR TOP OF FOOTING ELEVATIONS AND DETAILS FOR FOOTING STEP REQUIREMENTS.
7. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING THE EOR IMMEDIATELY OF ANY SPECIAL SOIL OR WATER CONDITIONS THAT EXIST ON SITE.
8. BACKFILL & COMPACTION SHALL BE INSPECTED AND TESTED BY A LICENSED GEOTECHNICAL ENGINEER OR QUALIFIED FIELD TECH. THE SUBMITTAL OF TESTING REPORTS SHALL BE PER SPECIAL INSPECTION REQUIREMENTS.
9. BACKFILL SHALL BE COMPACTED BY MECHANICAL MEANS. WATER INFILTRATION SHALL NOT BE ALLOWED. BACKFILL SHALL BE PLACED IN ALTERNATING LIFTS ON EACH SIDE OF THE FDN WALLS FOR STABILITY.
10. WALL FOOTINGS ARE TO BE CENTERED ON WALLS UNO PAD FOOTINGS ARE TO BE CENTERED ON COLUMNS UNO.
11. CONTRACTOR TO COORDINATE INTERIOR & EXTERIOR TOP OF FOOTINGS WITH MEP CONTRACTORS, PRIOR TO START OF CONSTRUCTION. PLUMBING TO BE ROUTED ABOVE FOOTINGS UNLESS APPROVED BY EOR. FOOTINGS MAY NEED TO BE LOCALLY LOWERED TO ACCOUNT FOR ADJACENT PLUMBING LINES OR BASINS THAT COULD UNDERMINE SUPPORTING SOIL ALONGSIDE OR BELOW FOOTINGS.
12. IF SHOWN ON FOUNDATION PLAN, DRAIN TILE IS FOR GRAPHICAL REPRESENTATION ONLY. SIZE AND LAYOUT TO BE CONFIRMED WITH MEP CONTRACTOR AND CIVIL DRAWINGS.

CONCRETE AND STEEL REINFORCEMENT NOTES:

1. CONCRETE AND STEEL REINFORCEMENT SHALL CONFORM TO AMERICAN CONCRETE INSTITUTION (ACI) CODES AND SPECIFICATIONS, LATEST EDITION.

ACI 301 "SPECIFICATIONS FOR STRUCTURAL CONCRETE"
ACI 315 "DETAILS & DETAILING OF CONCRETE REINFORCEMENT"
ACI 318 "BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE"
ACI 308R "COLD WEATHER CONCRETING"
2. CAST-IN-PLACE CONCRETE COMPRESSIVE STRENGTHS REQUIRED (28 DAY):

FOOTINGS 4000 PSI* PIERS / COLUMNS 4000 PSI
FOUNDATION WALLS 4000 PSI* INTERIOR SLABS 4000 PSI
ABOVE GRADE WALLS 4000 PSI* EXTERIOR SLABS 4500 PSI**
PRECAST TOPPING 4000 PSI*
*MAX W/C RATIO OF 0.55
**MAX W/C RATIO OF 0.45. MIX DESIGN TO BE AIR ENTRAINED PER ACI TABLE 19.3.3.1
3. SUBMIT CONCRETE MIX DESIGN & STRENGTH DATA TO EOR FOR APPROVAL. ALL ADMIXTURES ARE THE RESPONSIBILITY OF THE CONCRETE SUPPLIER'S ENGINEER.
4. CAST-IN-PLACE CONCRETE SHALL BE SUBJECT TO TESTING BY AN INDEPENDENT TESTING LABORATORY, SEE SPECS AND SPECIAL INSPECTION REQUIREMENTS.
5. ALL CONCRETE SHALL BE PLACED PER ACI & THOROUGHLY CONSOLIDATED BY MEANS OF A VIBRATOR, PARTICULARLY AROUND REINFORCEMENT STEEL AND CORNERS OF FORM WORK.
6. REINFORCING STEEL SHALL BE GRADE 60 DEFORMED, BILLET-STEEL, ASTM A615, UNO.
7. WELDED REINFORCING STEEL SHALL BE GRADE 60, LOW CARBON, ASTM A706, WHICH IS SPECIALLY MANUFACTURED TO BE WELDABLE.
8. WELDED WIRE FABRIC SHALL COMPLY WITH ASTM A82 AND A195 STANDARDS AND SHALL BE PLACED IN THE CENTER OF THE SLAB, UNO. LAP JOINTS A MINIMUM OF 6". EXTEND FABRIC TO BE WITHIN 1" OF SLAB EDGES.
9. PROVIDE ADEQUATE BOLSTERS, HIGH CHAIRS, SUPPORT BARS, ETC TO MAINTAIN THE SPECIFIED CLEARANCES FOR THE ENTIRE LENGTH OF ALL REINFORCING STEEL AND WELDED WIRE FABRIC.
10. PROVIDE EXTRA REINFORCEMENT AROUND ALL OPENINGS GREATER THAN 8" SQUARE OR ROUND. PROVIDE (2) #5 BARS @ 3' OC FOR EACH MAT OF BARS, AT EACH SIDE AND CORNER OF OPENING EXTENDING MINIMUM 18" PAST CORNER OF THE OPENING. PLACE 2" CLEAR FROM OPENING.
11. SEE DETAILS FOR REINFORCING LAP SPLICE SCHEDULE, UNO ON PLAN OR DETAILS.
12. CAST DOWELS, WITH STD 90 DEGREE HOOK, IN FOOTINGS FOR CONCRETE PIERS AND WALLS ABOVE. DOWELS SHALL BE THE SAME SIZE AND QTY AS THE VERTICAL REINFORCING (UNO).
13. EXTERIOR SLABS SHALL DRAIN FREELY AWAY FROM THE BUILDING. SEE CIVIL AND ARCH. DRAWINGS FOR ELEVATIONS AND SLOPES.
14. CONTROL SAWCUT JOINTS ARE TO BE EXECUTED AS SOON AS CONCRETE HAS HARDENED SUFFICIENTLY TO PREVENT AGGREGATE FROM DISLODGING BY SAW AND PRIOR TO SHRINKAGE STRESS CRACKING. SEE DETAIL 1/3301 FOR SLAB CONSTRUCTION JOINTS (CCJ) AND FOR SLAB CONTROL JOINTS (CJ), CONTRACTOR SHALL SUBMIT A PROPOSED JOINT LAYOUT TO ARCH/ENG FOR APPROVAL PRIOR TO SLAB PLACEMENT.

A. CONTROL JOINTS SHALL BE ON COLUMN LINES AND @ RE-ENTRANT CORNERS TO THE GREATEST EXTENT POSSIBLE WITH SPACING LESS THAN 12'-0" OC BETWEEN.
B. CONSTRUCTION JOINTS SHALL BE LOCATED SO AS NOT TO ALLOW A SINGLE SLAB POUR TO EXCEED 4000 SQUARE FEET UNLESS ALTERNATE MEASURES ARE TAKEN TO CONTROL SLAB CURLING & SHRINKAGE.
C. PROVIDE CJ OR CCJ JOINTS SO AS NOT TO EXCEED A SLAB UNIT ASPECT RATIO OF 15:1.
15. SYNTHETIC FIBERS, WHERE SPECIFIED ON PLAN FOR SLAB-ON-GRADE APPLICATIONS, TO BE MICROFIBER TYPE AND COMPLY WITH ASTM D7508 FOR USE IN PRODUCING TYPE III SYNTHETIC FIBER-REINFORCED CONCRETE MEETING THE REQUIREMENTS OF ASTM C1116 & ACI 544. MICROFIBER PERFORMANCE INTENT IS FOR REDUCTION OF PLASTIC SHRINKAGE CRACKS ONLY. MICROFIBERS ARE NOT INTENDED TO PROVIDE SECONDARY REINFORCEMENT FOR HARDENED CONCRETE CRACK CONTROL. HARDENED CONCRETE CRACK CONTROL TO BE ACCOMPLISHED BY USING ACI RECOMMENDED PRACTICES FOR MIX-DESIGN, PLACEMENT, CURING, AND CONTROL JOINT PLACEMENT. DOSAGE OF MICROFIBER TO BE DETERMINED BY CONCRETE MIX DESIGNER FOR THE SPECIFIC PROJECT APPLICATIONS. MINIMUM DOSAGE TO BE 1 SLB/CUBIC YARD. FIBER SELECTION AND CONCRETE PLACEMENT TO COMPLY WITH REQUIRED SLAB FINISHES IN PROJECT SPECIFICATIONS.
16. SEE ARCH DRAWINGS FOR DIMENSIONS OF STOOPS, FOUNDATION WALL HOLDOUTS, SLAB RECESSES, SLOPED SLABS & FOUNDATION WALL INSULATION. REFER TO ARCH DRAWINGS FOR BELOW GRADE VAPOR BARRIER REQUIREMENTS.

PRECAST/PRESTRESSED CONCRETE NOTES:

1. PRECAST/PRESTRESSED CONCRETE UNITS SHALL BE DESIGNED TO SUPPORT THE LOADS AND SPAN CONDITIONS PROVIDED ON THE CONTRACT DOCUMENTS. MEMBER DESIGN SHALL BE PERFORMED ACCORDING TO THE ACI 318 AND PCI LATEST EDITIONS AND UNDER THE SUPERVISION OF A PROFESSIONAL ENGINEER REGISTERED IN THE STATE OF THE PROJECT.
2. PRECAST TO PRECAST CONNECTIONS ARE THE RESPONSIBILITY OF PRECAST SUPPLIER. SHOW FIELD WELDS AND CONNECTION MATERIAL REQUIREMENTS ON SHOP DRAWING SUBMITTALS.
3. PRECAST/PRESTRESSED MEMBERS HAVE BEEN INDICATED ON THE DRAWINGS BY GENERAL SIZE AND DEPTH. THE STRUCTURAL DESIGN OF THESE MEMBERS AND THEIR LIFTING ACCESSORIES SHALL BE BY A PROFESSIONAL ENGINEER REGISTERED IN THE STATE OF THE PROJECT DELEGATED BY THE PRECAST MANUFACTURER.

A. DESIGN DEVIATIONS MUST BE SUBMITTED TO THE ARCHITECT & EOR FOR REVIEW AND APPROVAL PRIOR TO THE SUBMISSION OF STAMPED APPROVAL DRAWINGS AND CALCULATIONS.
B. DESIGN DEVIATIONS MUST PRODUCE AN INSTALLATION EQUIVALENT TO THE BASIC INTENT WITHOUT INCURRING ADDED COSTS.
4. SUBMITTAL DRAWINGS AND CALCULATIONS SHALL BE SIGNED AND STAMPED BY A PROFESSIONAL ENGINEER REGISTERED IN THE STATE OF THE PROJECT.
5. HEADERS AT OPENINGS THROUGH PRECAST MEMBERS SHALL BE SUPPLIED BY THE PRECAST SUPPLIER. ADJACENT PRECAST UNITS SHALL BE DESIGNED FOR THE LOADS AT EACH BEARING LOCATION.
6. PRECAST SUPPLIER SHALL BE RESPONSIBLE FOR OPENINGS 8" AND LARGER IN SIZE THROUGH PRECAST MEMBERS. HOLES LESS THAN 8" SHALL BE CUT BY THE TRADE WITH PRIOR APPROVAL OF THE PRECAST SUPPLIER.
7. PRECAST MEMBERS SHALL BE ERECTED ALTERNATELY ON EACH SIDE OF SUPPORTING WALLS AND BEAMS TO MAINTAIN STABILITY.
8. PRECAST SUPPLIER SHALL PROVIDE AND SHOP INSTALL EMBEDDED ITEMS IN PRECAST UNITS TO THE GREATEST EXTENT POSSIBLE. COORDINATE WITH APPROPRIATE TRADES.
9. CONTRACTOR IS RESPONSIBLE TO CONTACT THE PRECAST SUPPLIER TO DEVELOP AN APPROPRIATE FIELD CORRECTION WHEN FABRICATION OR FIELD INSTALLATION ERRORS RESULT IN MISALIGNMENT OF EMBEDS OR OTHER DEVIATIONS FROM THE APPROVED SHOP DRAWINGS. PROPOSED CORRECTION DETAILS WITH SUPPORTING CALCULATIONS MUST BE SUBMITTED TO THE EOR FOR APPROVAL PRIOR TO PERFORMING THE REPAIR.
10. PRECAST SUPPLIER SHALL VERIFY SIZE, QUANTITY, AND LOCATION OF OPENINGS WITH THE ARCHITECTURAL, STRUCTURAL, MECHANICAL, AND ELECTRICAL PLANS PRIOR TO FABRICATION OF THE PRECAST UNITS. THIS INCLUDES A FABRICATED CHASE FOR ELECTRICAL FEEDER LINES THROUGH PRECAST FLOORS.
11. PRECAST ASSEMBLIES TO SATISFY FIRE RATING REQUIREMENTS SPECIFIED BY ARCHITECTURAL DRAWINGS.
12. FOR PRECAST HORIZONTAL DIAPHRAGM APPLICATIONS, PRECAST SUPPLIER TO PROVIDE CONTINUITY ACROSS MEMBER JOINTS TO TRANSFER THE SPECIFIED DIAPHRAGM LOADS TO THE RESISTING VERTICAL SHEAR ELEMENT. IF FORCES EXCEED GROUTED KEYWAY CAPACITY, A MECHANICAL CONNECTION SHOULD BE DESIGNED AND SPECIFIED BY THE PRECAST ENGINEER. ALL GROUT SPECIFICATIONS SHOULD BE PROVIDED BY PRECAST SUPPLIER. HOT AND COLD WEATHER REQUIREMENTS TO BE SATISFIED FOR PREPARATION OF SUBSTRATE, PLACEMENT OF GROUT, AND CURING OF GROUT.

ALUMINUM FRAMING ASSEMBLIES NOTES:

1. THE DEFERRED ALUMINUM DESIGN SHALL CONFORM TO THE ALUMINUM DESIGN MANUAL (LATEST ADDITION). THIS DESIGN SHOULD BE DONE UNDER THE SUPERVISION OF A REGISTERED PROFESSIONAL ENGINEER IN THE STATE OF THE PROJECT.
2. THE ALUMINUM ASSEMBLIES SHALL BE DESIGNED FOR THE LIVE LOAD, SNOW LOAD, AND WIND LOAD (UPLIFT) LISTED UNDER THE DESIGN CRITERIA.
3. ALL CONNECTIONS TO THE BASE STRUCTURE SHALL BE FULLY DETAILED IN THE ALUMINUM APPROVAL SUBMITTAL AND CALCULATIONS SHALL BE PROVIDED. THE REACTIONS FROM THESE CONNECTIONS SHALL BE PROVIDED TO THE EOR TO CONFIRM THE CAPACITY OF THE SUPPORTING STRUCTURE.
4. A SEPARATION MATERIAL SHALL BE USED BETWEEN THE ALUMINUM ASSEMBLIES AND ANY DISSIMILAR METAL TO PREVENT CORROSION. THIS MATERIAL SHALL BE SHOWN ON THE SHOP DRAWINGS.
5. ALL SUBMITTAL CALCULATIONS SHALL BE SIGNED AND STAMPED BY A PROFESSIONAL ENGINEER REGISTERED IN THE STATE OF THE PROJECT.

POST INSTALLED ANCHORAGE NOTES:

1. POST INSTALLED ANCHORS NOTED ON PLAN AND/OR DETAILS NOTED SHALL BE AS FOLLOWS (UNO). IF ALTERNATIVE ANCHORS ARE DESIRED, CONTRACTOR MUST SUBMIT PRODUCT DATA FOR APPROVAL BY THE STRUCTURAL ENGINEER PRIOR TO ORDERING OF MATERIALS. ANCHORS USED TO TRANSFER STRUCTURAL LOADS MUST HAVE BEEN APPROVED BY METHODS OF ACI 318 APPENDIX D FOR MECHANICAL ANCHORS AND ICC-ES AC308 FOR ADHESIVE AND TORQUE-CONTROLLED ANCHORS. ADHESIVES USED IN COLD WEATHER MUST MEET ALL WEATHER REQUIREMENTS AND CODE REQUIREMENTS STATED ABOVE.

INJECTION ADHESIVE ANCHORS: SIMPSON SET-3G ADHESIVE.
A. THREADED RODS TO BE ASTM F1554 GRADE 55, UNO, CUT EMBEDDED END @ 45° ANGLE.
B. REBAR TO MATCH CONCRETE AND STEEL REINFORCEMENT NOTE REQUIREMENTS

EXPANSION ANCHORS: SIMPSON STRONG BOLT 2
SCREW ANCHORS: SIMPSON TITEN HD

POWDER ACTUATED FASTENERS (PAF): 0.157"Ø STEEL-TO-STEEL, THRU BASE METAL, 1/2" MIN. EDGE DISTANCE. 0.157"Øx11¼" STEEL TO CONCRETE & MASONRY. 3" MIN. CONCRETE EDGE DISTANCE (UNO).
2. POST INSTALLED ANCHORS SHALL ONLY BE USED WHERE SPECIFIED ON THE DRAWINGS. THE CONTRACTOR SHALL OBTAIN APPROVAL FROM THE STRUCTURAL ENGINEER PRIOR TO USING POST INSTALLED ANCHORS FOR MISSING OR MISPLACED CIP ANCHORS. CARE SHALL BE TAKEN TO AVOID CONFLICTS WITH EXISTING REINFORCING BARS. HOLES SHALL BE DRILLED AND CLEANED PER ANCHOR MANUFACTURER'S SPECIFICATIONS. ANCHORS AND ADHESIVE TO BE INSTALLED PER MANUFACTURER'S SPECIFICATIONS.

STEEL DECK:

1. STEEL DECK SHALL BE FABRICATED AND INSTALLED IN ACCORDANCE TO STEEL DECK INSTITUTE STANDARD (SDI).
2. STEEL DECK MANUFACTURER TO PROVIDE DETAILED SHOP DRAWINGS FOR REVIEW AND APPROVAL BY ARCHITECT AND STRUCTURAL ENGINEER.
3. ALTERNATE FASTENING OF STEEL DECK THAT MEETS OR EXCEEDS SPECIFIED FASTENING, MUST BE SUBMITTED TO ENGINEER FOR APPROVAL PRIOR TO WORK. SUBMITTAL TO INCLUDE FULL DESIGN DATA PREPARED BY MANUFACTURER COMPARING ALTERNATE, FASTENING LOAD VALUES TO SPECIFIED VALUES.
4. STEEL DECK SIZES (MIN. SIZE) UNO ON PLAN/DETAILS:

ROOF DECK - 1 1/2" TYPE B, 22 GA
FORM DECK - 1" TYPE C, 22 GA.
STOOP DECK - 1 1/2" TYPE C, 20 GA. (GALV).

ABBREVIATIONS AND SYMBOLS:

ALT	ALTERNATE	LSV	LONG SIDE VERTICAL
ALUM	ALUMINUM	LONG	LONGITUDINAL
AR	ANCHOR ROD	MAX	MAXIMUM
ARCH	ARCHITECT	MECH	MECHANICAL
BLDG	BUILDING	MEP	MECHANICAL/ELECTRICAL/PLUMBING
BO	BOTTOM OF	MIN	MINIMUM
BRG	BEARING	MISC	MISCELLANEOUS
CCJ	CONSTRUCTION CONTROL JOINT	NTS	NOT TO SCALE
CIP	CAST IN PLACE	NS	NON-SHRINK
CJ	CONTROL JOINT	OC	ON CENTER
CJP	COMPLETE JOINT PENETRATION	OD	OUTSIDE DIAMETER
CL	CENTER LINE	OF	OUTSIDE FACE
CLR	CLEAR	OH	OVERHEAD
CMU	CONCRETE MASONRY UNIT	OPP	OPPOSITE
CONC	CONCRETE	PAF	POWDER ACTUATED FASTENER
CONN	CONNECTION	PIC	PRECIPITATION
CONT	CONTINUOUS	PEMB	PRE-ENGINEERED METAL BUILDING
CONST	CONSTRUCTION	PLF	POUNDS PER LINEAR FOOT
COORD	COORDINATION	PSF	POUNDS PER SQUARE FOOT
DBE	DECK BEARING ELEVATION	PSI	POUNDS PER SQUARE INCH
DBL	DOUBLE	QTY	QUANTITY / QUANTITIES
DEG	DEGREE	RAD	RADIUS
DET	DETAIL	REF	REFERENCE
DIAG	DIAGONAL	REIN	REINFORCING / REINFORCE
DIM	DIMENSION	RET	RETURN
DL	DEAD LOAD	RO	ROUGH OPENING
DT	DOUBLE TEE	RTU	ROOF TOP UNIT
DWG	DRAWING	R/W	REINFORCE WITH
DWLR	DOWEL	SC	SLIP CRITICAL
EA	EACH	SCHED	SCHEDULE
EF	EACH FACE	SD	SEE DETAIL
EL	ELEVATION	SE	SQUARE FOOTAGE
EOR	ENGINEER OF RECORD	SF	SIMILAR
EQ	EQUAL	SIM	SLAB ON GRADE
(E)	EXISTING	SOG	SPECIFICATIONS
EXP	EXPANSION	SQ	SQUARE
EXT	EXTERIOR	SS	STAINLESS STEEL
EW	EACH WAY	STD	STANDARD
FAB	FABRICATOR	STF	STIFFENER
FDN	FOUNDATION	STRICT	STRUCTURAL
FT	FOOT	SW	SHEAR WALL
FTG	FOOTING	TEMP	TEMPORARY
GA	GAUGE	T&G	TOP & BOTTOM
GALV	GALVANIZE	T&G	TONGUE & GROOVE
GC	GENERAL CONTRACTOR	THK	THICK / THICKENED
GT	GIRDER TRUSS	TOB	TOP OF BEAM
HC	HOLLOW CORE	TOF	TOP OF FOOTING
HSA	HEADED STUD ANCHOR	TOJ	TOP OF JOIST
HORZ	HORIZONTAL	TOW	TOP OF WALL
HSS	HOLLOW STRUCTURAL SECTION	TOS	TOP OF STEEL
ICF	INSULATED CONCRETE FORM	TRANS	TRANSVERSE
ID	INSIDE DIAMETER	TYP	TYPICAL
IF	INSIDE FACE	ULT	ULTIMATE
INT	INTERIOR	UNO	UNLESS NOTED OTHERWISE
IT	INVERTED TEE	(V)	VERIFY
JBE	JOIST BEARING ELEVATION	VERT	VERTICAL
K	KIPS	W	WITH
KLF	KIPS PER LINEAR FOOT	W/O	WITHOUT
KSF	KIPS PER SQUARE FOOT	WP	WORK POINT
KSI	KIPS PER SQUARE INCH	WLF	WELDED WIRE FABRIC
LL	LIVE LOAD	Ø	DIAMETER
LLH	LONG LEG HORIZONTAL	@	AT
LSH	LONG SIDE HORIZONTAL	L	ANGLE
LLV	LONG LEG VERTICAL	⬇	ELEVATION



THIS PLAN, INCLUDING THE DESIGN AND CONCEPT, PREPARED BY SANDMAN STRUCTURAL ENGINEERS (SSE) AS INSTRUMENTS OF SERVICE SHALL REMAIN THE PROPERTY OF SSE AND ARE PROTECTED UNDER COPYRIGHT LAW. SSE SHALL RETAIN ALL COMMON LAW, STATUTORY, AND OTHER RESERVED RIGHTS, INCLUDING WITHOUT LIMITATION, THE COPYRIGHT THEREOF. UNAUTHORIZED USE IS STRICTLY PROHIBITED.

© 2024 BY SSE ALL RIGHTS RESERVED

Proj. Engineer: DT/GM
Drawn by: ML/BT
Date Issued: 06/27/2024

Revisions		DATE		COMMENTS	
#					

I hereby certify that this plan, specification, or report was prepared by me or under my direct supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Minnesota.